

Supplementary material

This document accompanies the manuscript *Non-monotonic source reversion across language-model pretraining checkpoints: behavioral trajectories and source-position residual interventions*. It contains the supplementary figures, the family-by-template semantic details, and the fact-cluster sensitivity analysis. Checkpoint rows are not independent statistical replicates. Machine-readable values and provenance are retained in the repository artifacts named below.

For semantic summaries, `analyze_semantic.py` clusters rows by the frozen `axis:category:subject:Y` key carried by each runner row and averages repeated rendered rows before its 4,000-draw bootstrap. The two deterministic key halves are robustness partitions used together in the gate, not independent held-out test sets.

The semantic preflight is a tokenizer/schema smoke check, not a model-based capability filter. It uses 16 items per axis and mode at one frozen revision, validates one-token X, Y, and Z answers and the single clean/corrupt token difference across templates A–C, and applies no prior-strength threshold or outcome-based item exclusion. The exact gate contract is recorded in `SEMANTIC-PREFLIGHT.md`.

Supplementary figures

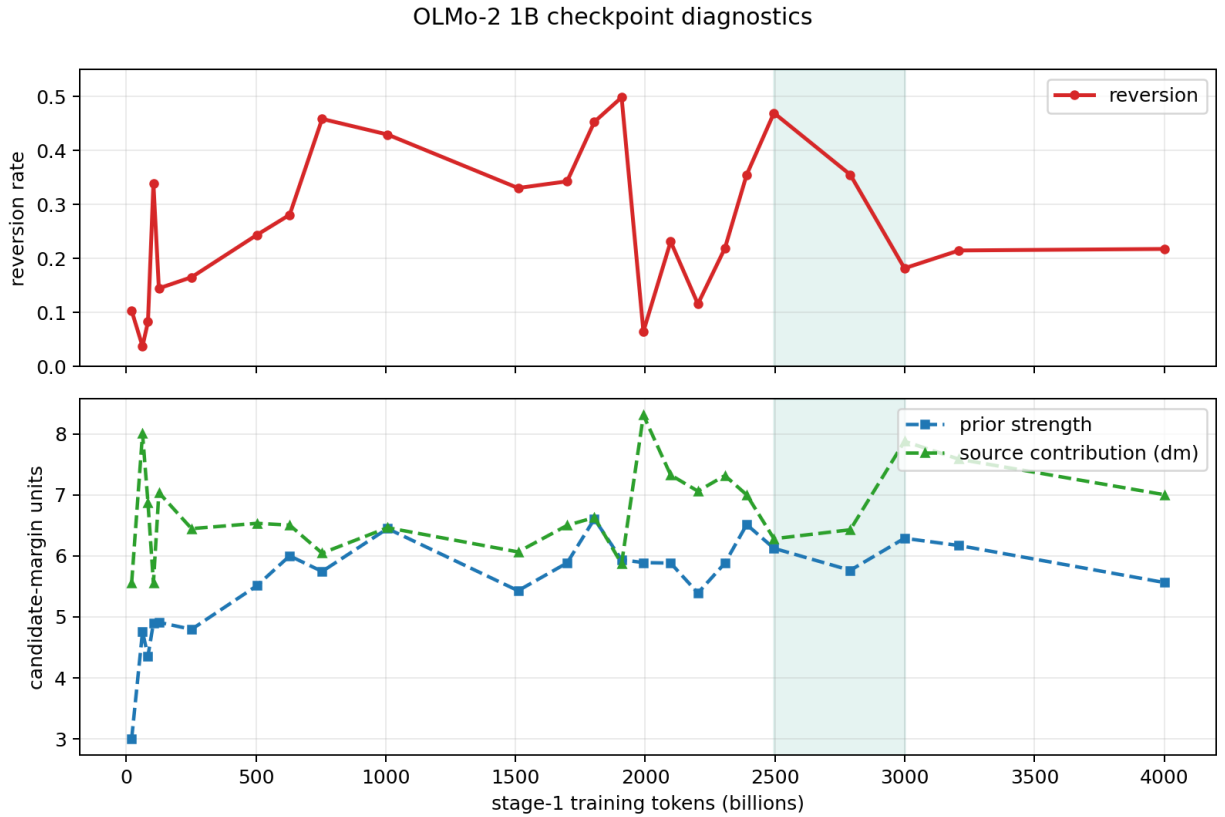


Figure S1. Reversion (top), source-erased prior strength, and source contribution across the OLMo-2 1B trajectory (bottom). The panels share the training-token axis; the lower panel uses candidate-margin units.

Candidate-choice decomposition of source reversion

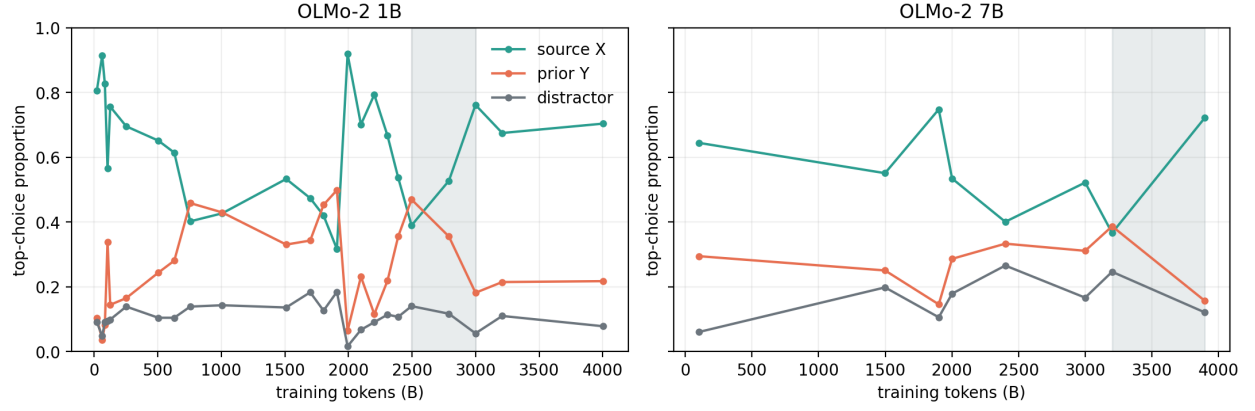


Figure S2. Candidate-choice decomposition across the complete OLMo-2 trajectories. Lines show item-cluster means for the explicit source, parametric prior, and distractors; shaded regions mark the selected late transitions.

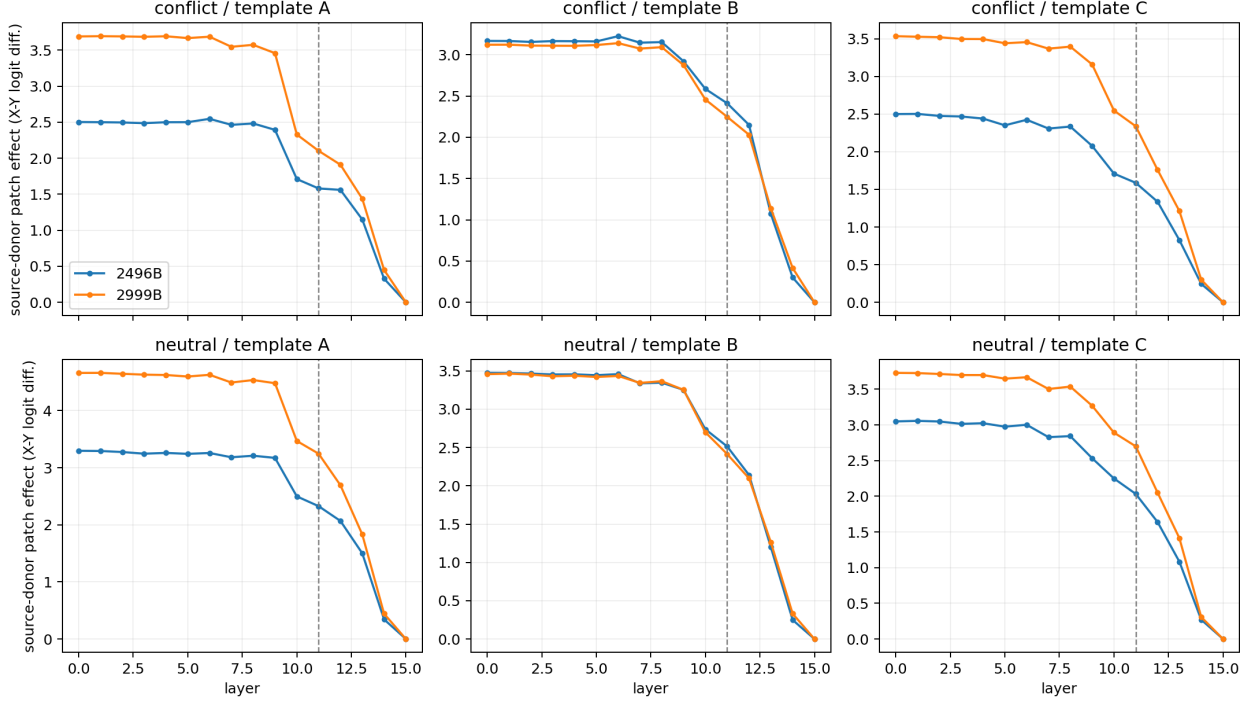


Figure S3. Layer-wise source-donor patch effect for conflict and neutral items at the OLMo-2 1B 2496B-to-2999B transition. The dashed line marks the first-75%-depth summary boundary.

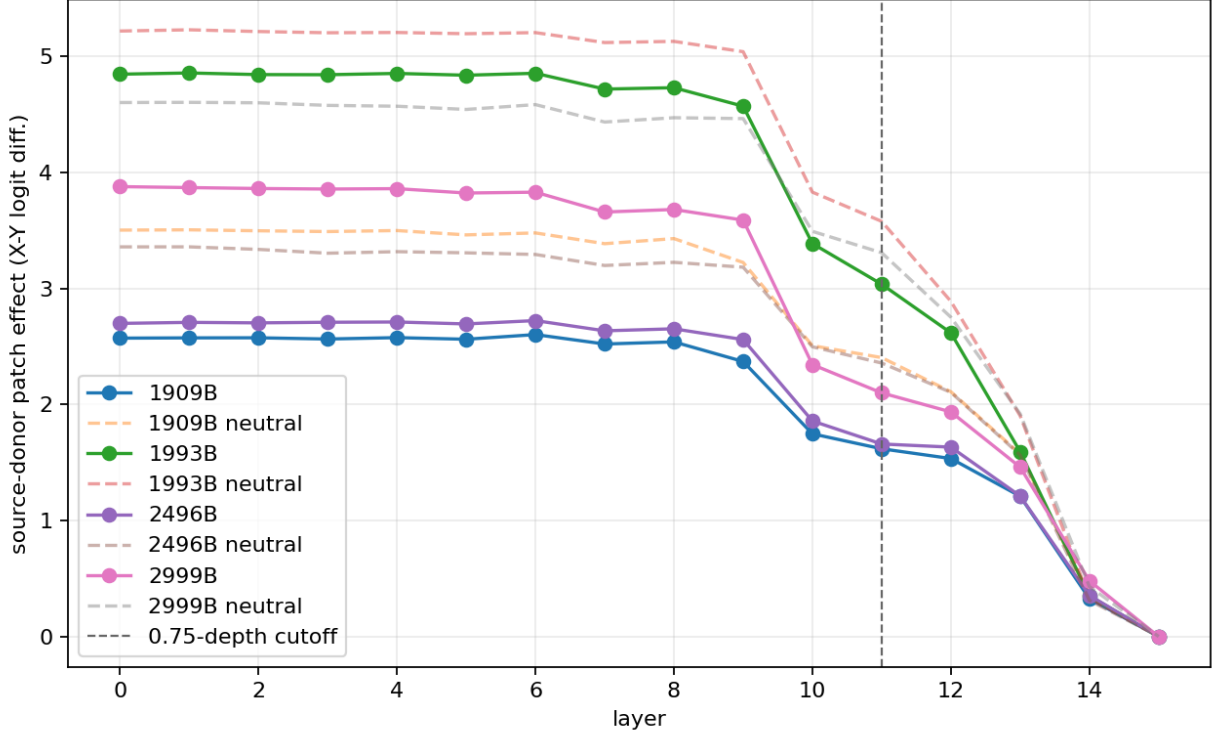


Figure S4. Layer-wise source-donor patch effect at the OLMo-2 1B global-maximum pair (1909B-to-1993B) and selected template-robust pair (2496B-to-2999B), including the neutral diagnostic.

Supplementary tables

Table S1. Family-by-template semantic peak details. These conditional descriptive summaries underlie Table 5 and Figure 2. Intervals and means are copied from `cross_family_semantic_pool.json`; no rows were removed after observing the effect direction.

Behavior: source gain

Source gain is the clean-minus-corrupt X-minus-Y margin in single-token logit-difference units. The `reversion` column is the pairwise Y-over-X error mean retained by the semantic summary.

Family	Axis	Template	Source gain mean	95% CI	Reversion mean
Amber-7B	Type/category	A	4.462	[4.462, 4.462]	0.000
Amber-7B	Type/category	B	3.618	[3.618, 3.618]	0.073
Amber-7B	Type/category	C	2.198	[2.198, 2.198]	0.178
Amber-7B	Relation/slot	A	4.700	[4.700, 4.700]	0.195
Amber-7B	Relation/slot	B	4.990	[4.990, 4.990]	0.043
Amber-7B	Relation/slot	C	3.061	[3.061, 3.061]	0.027
Amber-7B	Naturalistic	A	4.905	[4.603, 5.209]	0.082
Amber-7B	Naturalistic	B	3.858	[3.708, 4.005]	0.038
Amber-7B	Naturalistic	C	2.925	[2.795, 3.047]	0.006
Pythia-6.9B	Type/category	A	3.120	[2.848, 3.394]	0.161
Pythia-6.9B	Type/category	B	2.484	[2.275, 2.699]	0.306
Pythia-6.9B	Type/category	C	1.494	[1.314, 1.675]	0.463
Pythia-6.9B	Relation/slot	A	2.233	[1.930, 2.515]	0.913
Pythia-6.9B	Relation/slot	B	3.445	[2.351, 4.527]	0.196

Family	Axis	Template	Source gain mean	95% CI	Reversion mean
Pythia-6.9B	Relation/slot	C	1.590	[1.358, 1.820]	0.695
Pythia-6.9B	Naturalistic	A	4.403	[3.913, 4.881]	0.031
Pythia-6.9B	Naturalistic	B	2.702	[2.419, 3.001]	0.300
Pythia-6.9B	Naturalistic	C	3.660	[3.403, 3.921]	0.012

The degenerate intervals for some Amber type/relation rows are a property of the stored summary and should not be read as evidence of zero sampling uncertainty. The machine-readable artifact is the source of record.

Source-donor patch effect

The patch effect is the item-wise maximum layer shift from the corrupt baseline, averaged within each template. The selected layer is zero-indexed in the family-specific residual output.

Family	Axis	Template	Patch effect mean	95% CI	Selected layer
Amber-7B	Type/category	A	4.892	[4.483, 5.308]	5
Amber-7B	Type/category	B	4.075	[3.675, 4.500]	7
Amber-7B	Type/category	C	2.583	[2.267, 2.933]	10
Amber-7B	Relation/slot	A	4.992	[4.725, 5.258]	18
Amber-7B	Relation/slot	B	5.333	[5.117, 5.558]	12
Amber-7B	Relation/slot	C	3.300	[3.058, 3.542]	18
Amber-7B	Naturalistic	A	5.442	[5.042, 5.842]	12
Amber-7B	Naturalistic	B	4.050	[3.783, 4.325]	0
Amber-7B	Naturalistic	C	3.133	[2.925, 3.342]	5
Pythia-6.9B	Type/category	A	3.273	[2.961, 3.603]	3
Pythia-6.9B	Type/category	B	2.711	[2.477, 2.948]	3
Pythia-6.9B	Type/category	C	1.971	[1.739, 2.222]	2
Pythia-6.9B	Relation/slot	A	2.541	[2.219, 2.849]	5
Pythia-6.9B	Relation/slot	B	3.548	[3.001, 4.105]	0
Pythia-6.9B	Relation/slot	C	1.880	[1.619, 2.138]	17
Pythia-6.9B	Naturalistic	A	4.436	[4.073, 4.823]	7
Pythia-6.9B	Naturalistic	B	2.854	[2.636, 3.072]	0
Pythia-6.9B	Naturalistic	C	3.677	[3.430, 3.907]	3

The pooled means in Table 5 apply equal template weight within each family and then equal family weight; they are not obtained by pooling the 18 rows as if they were independent observations.

Table S2. Fact-cluster sensitivity. The existing rows are reweighted so that each (category, subject, Y) tuple contributes one observation. Estimates are supplementary sensitivity summaries, not replacements for the primary item-cluster analysis; intervals use 10,000 fact-cluster bootstrap draws. The patch columns use single-token X-minus-Y logit-difference units.

Model	Fact clusters	Excursion	95% CI	Extrema (B)
OLMo-2 1B	28	0.482	[0.380, 0.607]	63 / 1909 / 1993
OLMo-2 7B	28	0.242	[0.161, 0.321]	1901 / 3201 / 3896

The fact-weighted patch summaries are shown below. The table reports fact-weighted absolute early-depth patch effects (q_{early} , q_{late}) and their change ($\text{delta}_q = q_{\text{late}} - q_{\text{early}}$); intervals are fact-cluster bootstrap 95% intervals.

Group	Matched rows	Facts	q_early [95% CI]	q_late [95% CI]	delta_q [95% CI]
1B conflict A	116	23	2.367 [2.004, 2.726]	3.423 [3.053, 3.782]	+1.056 [0.826, 1.292]
1B conflict B	116	23	3.204 [2.876, 3.557]	3.049 [2.865, 3.224]	-0.155 [-0.495, 0.179]
1B conflict C	116	23	2.213 [1.837, 2.627]	3.322 [2.886, 3.781]	+1.109 [0.783, 1.438]
1B neutral A	120	70	3.070 [2.794, 3.343]	4.382 [4.111, 4.651]	+1.311 [1.111, 1.517]
1B neutral B	120	70	3.455 [3.150, 3.770]	3.256 [3.043, 3.475]	-0.199 [-0.444, 0.040]
1B neutral C	120	70	2.750 [2.445, 3.053]	3.559 [3.270, 3.857]	+0.809 [0.538, 1.093]
7B conflict A	57	21	0.767 [0.518, 0.978]	1.764 [1.502, 2.017]	+0.997 [0.754, 1.239]
7B conflict B	57	21	1.964 [1.714, 2.199]	2.619 [2.261, 2.957]	+0.656 [0.409, 0.900]
7B neutral A	60	46	1.327 [1.084, 1.576]	2.630 [2.262, 3.022]	+1.303 [1.014, 1.611]
7B neutral B	60	46	2.223 [1.980, 2.477]	2.764 [2.434, 3.092]	+0.541 [0.334, 0.742]

Reproducibility note

The complete semantic detail table is also available as `TABLE-S1-SEMANTIC-DETAILS.md`; the machine-readable fact sensitivity output and analysis definition are in `CLUSTER-SENSITIVITY.md` and `cluster_sensitivity.json`. The checksum and container provenance are recorded in `SUPPLEMENTARY-REPRODUCIBILITY-NOTE.md`. Raw semantic rows are retained in the experiment archive but are not yet deposited in a persistent public archive.